A GPS navigation system works by analysing radio signals received from GPS satellites. A GPS receiver provides its current location 24x7. GPS systems were traditionally used as a vital defence technology and later used for commercial purposes. In India, GPS technology has been used for advanced public transport systems. Bangalore Metropolitan Transport Corporation (BMTC) was the first to deploy off-line GPS technology on its vehicles on an experimental basis and after realising the benefits, commissioned a project to implement real time GPS for its fleet to track and provide passengers with real time departure and arrival times of buses.

A similar model is being adopted by private cab operators. Tracking vehicles through GPS not only leads to significant savings by instructing the nearest cab to pick up a customer, but also adds to customer delight due to shortened pick-up times.

Companies use GPS for fleet management broadly to track the exact location of the vehicles during the course of the day, the distance traveled and the locations visited and any instances of over speeding. Such systems can also be utilised by companies to validate the claims of their sales personnel.

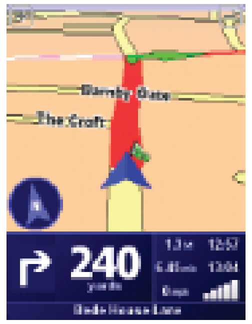
GPS receivers, once the prerogative of commercial users, are now cheap enough to be affordable to almost anyone — prices have fallen to as little as Rs. 2,500.

As GPS gets integrated with basic mobile phones, its usage in our daily lives will increase as well.

While a mobile phone has become a basic necessity these days, and has moved beyond being just a means of voice communication, a mobile phone with inbuilt GPS capabilities can bring unlimited possibilities.

Navigation

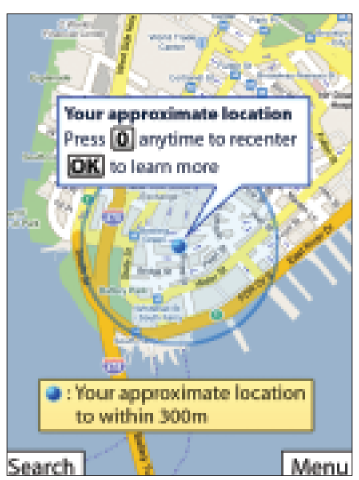
GPS navigation systems provide useful information about an individual or a vehicle's current position, regardless of the vehicle travelling on road, water or air. Modern day mobile phones with GPS antennas can work as navigation systems and run mapping software that can track the current position of the vehicle and provide turn-by-turn driving instructions as well. Google Maps, a mapping software that has gained popularity globally, can be used to determine the current position on a map. Google Maps is a simple, yet powerful application, which can work on most mobile phones and provide precise location (if the phone has an internal or external GPS antenna) or approximate location — if the phone does not have a GPS antenna. Google Maps also has capability to provide traffic density information (which can help choose an optimal route) and routing information (turn-by-turn driving/walking directions). Unfortunately, the traffic information feed and routing information is not available for India, yet.



Nav4All in action

TeleNav giving you precise directions

Google Maps helping you find your way



There are a number of other alternative systems, which provide navigation and routing information. Airtel provides an application (Wayfinder) for GPS-enabled Blackberry users (Blackberry 8800, 8310, Bold, etc.), which offers voice-enabled turn-by-turn routing information. The maps for this application have been provided by *MapMyIndia*, which has fairly precise maps and routing information for may Indian cities. With this application on a Blackberry handheld, one can get around most places in India quite easily. The software also provides route summary, which includes total distance from start point to destination, distance left, time taken, etc. The system can also predict the expected arrival time with reasonable accuracy. GPS-enabled Nokia phone users can also use a similar application for navigation. The other similar systems are Nav4All, TeleNav, etc.

Geo tagging

Geo tagging is another interesting application, which extends the capability of a camera or mobile phone, that has inbuilt GPS. Standard digital cameras have the capability to tag pictures with basic information such as date, time and camera settings. A GPS-enabled camera (or camera phone) can add location information as an additional tag to each picture taken. This tag can describe the exact location of the picture on our planet. This functionality helps users to keep track of the places they have visited, without having to manually enter such information.

Anti-theft devices

Today, several security systems are designed by embedding GPS units in vehicles. These units when integrated with automobile security systems can notify the vehicle owner by phone or email if the vehicle security alarms are triggered, and help in identifying the exact location of the vehicle.

Sports

Runners, cyclists and other health conscious individuals can use GPS-enabled mobile phones to monitor their speed and distance covered. The famous French cycling event *Tour de France* has riders equipped with GPS devices that transmit their speed and location at all times. GPS systems are also used to capture player work rates and training loads to monitor and prescribe exercise and recovery routines. Modern day golf courses also install GPS units on golf carts, and these are being used to monitor among other things, the distance a ball has been hit and the distance to the green.

Recreational activities

Trekkers and hikers use GPS-enabled mobile phones to create a track log of the hike. GPS has also given rise to new games such as *Geocache*, which combine the physical and virtual worlds. The game, played across the globe by adventure seekers equipped with GPS devices involves locating and hiding containers called geocaches (water proof containers containing a logbook, toys and trinkets of low value) outdoors and sharing experiences online with the player community. GPS is also finding its place in fishing activities. Fishing in large water bodies makes it difficult to keep a track of the location where success was met. GPS devices help by marking such locations, allowing you to return to them when required.

What lies ahead

The convergence of GPS with mobile phones brings additional opportunities for people to use this data to share their location information with their friends. The location-based information can also be published in the individual's profile on social networking sites like Facebook and Linkedin. The same application can also be used by parents to track their kids on a real-time basis.

The GPS system is a testimonial of how man's motivations and desires to probe his own curiosities have helped him create a technological marvel. GPS systems have triggered technological fantasies around the globe and applications of this technology appear to be virtually unlimited. India is slowly beginning to embrace this technology to answer some of the many challenges facing a burgeoning population and we shall soon begin to see the penetration rise.

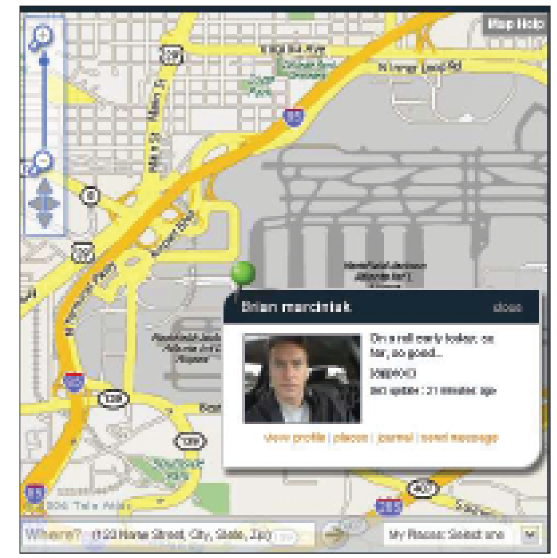
Convergence of GPS and mobile phones — going beyond navigation

The initial usage of a GPS-enabled mobile phone was to simply track one's location on a mapping application like Google Maps. Soon after, came the navigation systems, which helped users get from one point to another by assisting them with turn-by-turn driving directions. Mobile navigation systems are good, but are limited in their capabilities. Also, the usage of these applications is quite limited. Since navigation systems are used only when one travels to unknown places, initial research indicates that a typical user does not use these systems more than once in two weeks. Such low usage may not justify installing a rather heavy application just for the purpose of navigation.

The next set of applications for GPS-enabled mobile phones relies on the fact that people would like to do much more with the real-time location information available to them. One of the most fascinating applications shares this information with a network of trusted friends. This is known as Buddy Tracking System. A number of systems are available that help you share their location information in real time with their friends. The possibilities multiply manifold if the location information can be shared through a number of different media platforms. Some of the advanced applications can integrate one's location information on a phone, web and other platforms. Therefore, a user may choose to share his location information with all or some of his friends that he or she is connected to on social networking sites such as Orkut, Facebook or LinkedIn.

Wayfinder: the name says it all

One such application is Loopt (**www.loopt.com**), which enables buddy tracking using a GPS-enabled mobile phone and can publish this information across a range of platforms. *Loopt* is a great buddy-tracking system, but it is linked to the telecom operator, which implies that its usage is limited to only those users whose telecom service provider provides this service. Unfortunately, this service is not available with any telecom service provider in India.



Loopt can tell your friends where you are at any given time

Another interesting application is Whrrl (**www.whrrl.com**), which provides buddy tracking service, and in addition, offers event capture. Essentially, users of Whrrl can record their comments / feedback about the place (restaurant, club, etc) they visit, and these comments will be visible to their

Whrrl can tell your buddies about that wonderful restaurant you visited last night